RepairDesk — Complete Solution

> The whole project, every file. Use it to **check your work**, not to skip the thinking. Build from the story first; read the hints if you get stuck. All of this code runs.

The final layout

```
<repo folder>/
├── data/
│   └── sample/
│       └── messages.json
├── scripts/
│   └── intake_samples.py
├── src/
│   ├── agents/
│   │   ├── __init__.py
│   │   ├── concierge.py
│   │   ├── intake.py
│   │   └── router.py
│   ├── models/
│   │   ├── __init__.py
│   │   └── llm.py
│   ├── prompts/
│   │   ├── concierge.txt
│   │   ├── intake.txt
│   │   └── router.txt
│   ├── services/
│   │   ├── __init__.py
│   │   └── shop_service.py
│   ├── utils/
│   │   ├── __init__.py
│   │   ├── history.py
│   │   ├── logger.py
│   │   └── prompts.py
│   ├── __init__.py
│   ├── config.py
│   ├── exceptions.py
│   └── schemas.py
├── tests/
│   ├── integration/
```

```
│   │   └── test_live_agents.py
│   └── unit/
│       ├── test_config.py
│       ├── test_history.py
│       └── test_schemas.py
├── .env.example
├── main.py
└── requirements.txt
```

The project files you make once

These are the plain setup files. Copy them exactly — nothing here comes from another project.

`requirements.txt`

```
openai
python-dotenv
pydantic
```

`.env.example` (copy it to `.env` and put your real key in `.env`)

```
OPENAI_API_KEY=sk-your-key-here
LOG_LEVEL=INFO
```

`.gitignore`

```
.env
.venv/
__pycache__/
*.pyc
```

Every folder under `src/`, and `src/` itself, needs an **empty** `__init__.py` so Python treats it as a package: `src/__init__.py`, `src/agents/__init__.py`, `src/models/__init__.py`, `src/services/__init__.py`, `src/utils/__init__.py`. They hold nothing — the file just needs to exist.

`LICENSE` (MIT — fill in the year and your name)

```
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```

```
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OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
SOFTWARE.
```

Settings, errors, logging

```
# src/exceptions.py
# The project's named errors. MissingConfigError: a required setting is
# missing from .env.
class MissingConfigError(Exception):
    """Raised when a required setting is missing from the environment."""
    pass
```

```
# src/config.py
# Loads the settings from .env into a checked Config object. load_config()
# raises MissingConfigError when OPENAI_API_KEY is missing.
import os
from dataclasses import dataclass
from dotenv import load_dotenv
from src.exceptions import MissingConfigError

@dataclass
class Config:
    openai_api_key: str
    log_level: str

def require_env(key: str) -> str:
    value = os.getenv(key)
    if value is None or value == "":
        raise MissingConfigError(
```

```
            f"Required environment variable is missing: {key}"
        )
    return value

def load_config() -> Config:
    load_dotenv()
    api_key = require_env("OPENAI_API_KEY")
    log_level = os.getenv("LOG_LEVEL", "INFO")
    return Config(openai_api_key=api_key, log_level=log_level)
```

```
# src/utils/logger.py
# get_logger(name): one logging setup for every file — a timestamp, the
# file name and the level on every line.
import logging
import os

def get_logger(name: str) -> logging.Logger:
    logger = logging.getLogger(name)
    if not logger.handlers:
        level_name = os.getenv("LOG_LEVEL", "INFO")
        level = getattr(logging, level_name.upper(), logging.INFO)
        handler = logging.StreamHandler()
        handler.setLevel(level)
        log_format = "%(asctime)s %(name)s %(levelname)s %(message)s"
        handler.setFormatter(logging.Formatter(log_format))
        logger.addHandler(handler)
        logger.setLevel(level)
    return logger
```

Prompts and history helpers

```
# src/utils/prompts.py
# load_prompt(name): reads src/prompts/<name>.txt, so prompts live in text
# files instead of Python strings.

def load_prompt(name: str) -> str:
    with open("src/prompts" + "/" + name + ".txt") as prompt_file:
        return prompt_file.read().strip()
```

```
# src/utils/history.py
# Keeps a chat under the model's limit: estimate_tokens() guesses its size
# (about 4 characters per token), and trim_history() drops the oldest turns
# — never the system prompt.
```

```
def estimate_tokens(history: list) -> int:
    total = 0
    for message in history:
        total = total + len(message["content"])
    return total // 4

def trim_history(history: list, max_tokens: int) -> list:
    while estimate_tokens(history) > max_tokens and len(history) > 2:
        del history[1]
    return history
```

The one file that talks to OpenAI

```
# src/models/llm.py
# The only file that talks to the OpenAI SDK: create_client(), the startup
# key check, send_message(), and stream_message() for live replies.
import openai
from openai import OpenAI

from src.config import Config

MODEL = "gpt-4o-mini"

def create_client(config: Config) -> OpenAI:
    return OpenAI(api_key=config.openai_api_key)

def check_api_key_at_startup(client: OpenAI) -> bool:
    try:
        client.chat.completions.create(
            model=MODEL,
            messages=[{"role": "user", "content": "ping"}],
            max_tokens=1,
        )
    except openai.AuthenticationError:
        return False
    return True

def send_message(client: OpenAI, history: list, user_input: str) -> str:
    history.append({"role": "user", "content": user_input})
    response = client.chat.completions.create(model=MODEL, messages=history)
    reply = response.choices[0].message.content
    history.append({"role": "assistant", "content": reply})
    return reply
```

```
def stream_message(client: OpenAI, history: list, user_input: str) -> str:
    history.append({"role": "user", "content": user_input})
    stream = client.chat.completions.create(
        model=MODEL, messages=history, stream=True
    )
    full_reply = ""
    for chunk in stream:
        piece = chunk.choices[0].delta.content
        if piece:
            print(piece, end="", flush=True)
            full_reply = full_reply + piece
    print()
    history.append({"role": "assistant", "content": full_reply})
    return full_reply
```

The fixed shapes

```
# src/schemas.py
# The shapes the model must answer in. RepairTicket: one repair ticket.
# RouteDecision: which agent gets a message, and why.
from typing import Literal
from pydantic import BaseModel

class RepairTicket(BaseModel):
    customer_name: str | None
    device: Literal["phone", "laptop", "tablet", "other"]
    urgency: Literal["low", "medium", "high"]
    summary: str

class RouteDecision(BaseModel):
    agent: Literal["concierge", "intake"]
    reason: str
```

The three agents

```
# src/agents/intake.py
# The Intake agent: turns one problem message into a RepairTicket, or
# returns None when the model's answer can't be used.
from openai import OpenAI
from pydantic import ValidationError

from src.models.llm import MODEL
```

```
from src.schemas import RepairTicket
from src.utils.logger import get_logger
from src.utils.prompts import load_prompt

def intake_agent(client: OpenAI, message: str):
    logger = get_logger(__name__)
    messages = [
        {"role": "system", "content": load_prompt("intake")},
        {"role": "user", "content": message},
    ]
    try:
        completion = client.beta.chat.completions.parse(
            model=MODEL, messages=messages, response_format=RepairTicket
        )
    except ValidationError as e:
        logger.warning(f"Intake output did not match RepairTicket: {e}")
        return None
    reply = completion.choices[0].message
    if reply.refusal:
        logger.warning(f"Intake agent refused: {reply.refusal}")
        return None
    return reply.parsed
```

```
# src/agents/concierge.py
# The Concierge agent: start_history() begins a chat with its system
# prompt, and concierge_agent() trims the chat and streams the reply.
from openai import OpenAI

from src.models.llm import stream_message
from src.utils.history import trim_history
from src.utils.prompts import load_prompt

def start_history() -> list:
    return [{"role": "system", "content": load_prompt("concierge")}]

def concierge_agent(client: OpenAI, history: list, message: str) -> str:
    trim_history(history, 3000)
    return stream_message(client, history, message)
```

```
# src/agents/router.py
# The router: decide() asks the model which agent should answer a message,
# and falls back to the Concierge when the answer can't be used.
from openai import OpenAI
from pydantic import ValidationError
```

```
from src.models.llm import MODEL
from src.schemas import RouteDecision
from src.utils.logger import get_logger
from src.utils.prompts import load_prompt

def decide(client: OpenAI, message: str) -> RouteDecision:
    logger = get_logger(__name__)
    messages = [
        {"role": "system", "content": load_prompt("router")},
        {"role": "user", "content": message},
    ]
    decision = RouteDecision(
        agent="concierge", reason="router could not decide"
    )
    try:
        completion = client.beta.chat.completions.parse(
            model=MODEL, messages=messages, response_format=RouteDecision
        )
        reply = completion.choices[0].message
        if reply.parsed is not None:
            decision = reply.parsed
    except ValidationError as e:
        logger.warning(f"Router output did not match RouteDecision: {e}")
    logger.info(f"routed to {decision.agent} ({decision.reason}): {message}")
    return decision
```

The service that ties it together

```
# src/services/shop_service.py
# The app's flow, apart from the keyboard: handle_message() asks the router,
# then calls the chosen agent. print_ticket() shows a ticket.
from openai import OpenAI

from src.agents.concierge import concierge_agent
from src.agents.intake import intake_agent
from src.agents.router import decide
from src.schemas import RepairTicket
from src.utils.logger import get_logger

def print_ticket(ticket: RepairTicket) -> None:
    name = ticket.customer_name
    if name is None:
        name = "(not given)"
    print("Repair ticket created:")
    print(f"  customer: {name}")
```

```
    print(f"  device:   {ticket.device}")
    print(f"  urgency:  {ticket.urgency}")
    print(f"  summary:  {ticket.summary}")

def handle_message(client: OpenAI, history: list, message: str) -> None:
    logger = get_logger(__name__)
    decision = decide(client, message)
    if decision.agent == "intake":
        ticket = intake_agent(client, message)
        if ticket is None:
            print("Sorry, I couldn't turn that into a repair ticket — please")
            print("describe the problem again in a little more detail.")
        else:
            logger.info(f"ticket: {ticket.device} / {ticket.urgency}")
            print_ticket(ticket)
    else:
        print("Concierge: ", end="", flush=True)
        concierge_agent(client, history, message)
```

The prompts

`src/prompts/concierge.txt`

```
You are the RepairDesk concierge. Be friendly and brief.
Answer general questions about the repair shop: hours, prices, services.
```

`src/prompts/intake.txt`

```
You take repair intake. Turn the customer message into a repair ticket.
Fill customer_name only if the customer states their name - otherwise leave
it null, never guess one. Pick device from the message. Use urgency "high"
only if the customer says it is urgent or needed today.
```

`src/prompts/router.txt`

```
You route a message to one agent. Use "intake" if the customer describes
a device problem to repair. Use "concierge" for general questions, prices,
hours or small talk. Give a one-sentence reason.
```

The sample data, the runner, and main.py

`data/sample/messages.json`

```
{
  "intake": [
    "Hi, I'm Sara Khan. My phone screen is cracked, please fix it today.",
    "my laptop won't turn on, tried charging overnight, still nothing",
    "my tablet is really slow and freezes a lot, no rush"
  ],
  "routing": [
    {"message": "hi there, how are you today?", "expected": "concierge"},
    {"message": "what are your repair hours?", "expected": "concierge"},
    {"message": "my phone screen is cracked, fix it", "expected": "intake"},
    {"message": "my laptop won't turn on at all", "expected": "intake"}
  ]
}
```

```
# scripts/intake_samples.py
# Runs the Intake agent on every sample message and prints each ticket.
# Run from the repo root: python -m scripts.intake_samples
import json

from src.agents.intake import intake_agent
from src.config import load_config
from src.models.llm import create_client

SAMPLES_PATH = "data/sample/messages.json"

def main() -> None:
    config = load_config()
    client = create_client(config)
    with open(SAMPLES_PATH) as samples_file:
        samples = json.loads(samples_file.read())
    for message in samples["intake"]:
        print(f"\nMessage: {message}")
        ticket = intake_agent(client, message)
        if ticket is None:
            print("No ticket — the Intake agent couldn't extract one.")
        else:
            print(f"Ticket:  {ticket}")

if __name__ == "__main__":
    main()
```

```
# main.py
# The file you run: python main.py
```

```
# Startup, the input loop and the error messages — every message goes to
# handle_message(), which picks the agent.
import sys

import openai

from src.agents.concierge import start_history
from src.config import load_config
from src.exceptions import MissingConfigError
from src.models.llm import check_api_key_at_startup, create_client
from src.services.shop_service import handle_message
from src.utils.logger import get_logger

def main() -> None:
    try:
        config = load_config()
    except MissingConfigError as e:
        print(f"Setup problem: {e}")
        sys.exit(1)

    logger = get_logger(__name__)
    client = create_client(config)
    if not check_api_key_at_startup(client):
        logger.error("Authentication failed — check OPENAI_API_KEY.")
        print("Login failed — check OPENAI_API_KEY in your .env file.")
        sys.exit(1)

    history = start_history()
    print("RepairDesk AI — type a message, or /quit to exit.")

    while True:
        message = input("You: ").strip()
        if message == "/quit":
            break
        if message == "":
            continue
        try:
            handle_message(client, history, message)
        except openai.AuthenticationError:
            logger.error("API key stopped working mid-session.")
            print("Login failed — check OPENAI_API_KEY in your .env file.")
            sys.exit(1)
        except openai.RateLimitError:
            logger.warning("Rate limited after the client's retries.")
            print("We're busy right now — wait a moment and try again.")
        except openai.BadRequestError as e:
            logger.warning(f"Request rejected (likely too long): {e}")
```

```
            if history[-1]["role"] == "user":
                history.pop()
            print("That message was too long to handle — please shorten it.")

if __name__ == "__main__":
    main()
```

The tests

```
# tests/unit/test_config.py
# Checks the settings code with no API call. Each case prints OK or WRONG.
# It only changes settings inside this test run — your .env is never touched.
# Run from the repo root: python -m tests.unit.test_config
import os

from src.config import load_config, require_env
from src.exceptions import MissingConfigError

TEST_SETTING = "REPAIRDESK_TEST_SETTING"

def test_missing_setting() -> None:
    os.environ.pop(TEST_SETTING, None)
    result = "WRONG"
    try:
        require_env(TEST_SETTING)
    except MissingConfigError:
        result = "OK"
    print(f"case 1 (missing setting) -> {result}")

def test_empty_setting() -> None:
    os.environ[TEST_SETTING] = ""
    result = "WRONG"
    try:
        require_env(TEST_SETTING)
    except MissingConfigError:
        result = "OK"
    print(f"case 2 (empty setting)   -> {result}")

def test_config_loads() -> None:
    os.environ["OPENAI_API_KEY"] = "sk-test-123"
    config = load_config()
    result = "OK"
    if config.openai_api_key != "sk-test-123":
        result = "WRONG"
    print(f"case 3 (config loads)    -> {result}")
```

```
if __name__ == "__main__":
    test_missing_setting()
    test_empty_setting()
    test_config_loads()
```

```
# tests/unit/test_history.py
# Checks the history helpers. Each case prints OK or WRONG. No API key.
# Run from the repo root: python -m tests.unit.test_history
from src.utils.history import estimate_tokens, trim_history

def test_estimate_tokens() -> None:
    history = [
        {"role": "system", "content": "a" * 20},
        {"role": "user", "content": "b" * 20},
    ]
    result = "OK"
    if estimate_tokens(history) != 10:
        result = "WRONG"
    print(f"case 1 (40 chars = 10 tokens) -> {result}")

def test_short_history_untouched() -> None:
    history = [
        {"role": "system", "content": "be nice"},
        {"role": "user", "content": "hi"},
    ]
    trim_history(history, 100)
    result = "OK"
    if len(history) != 2:
        result = "WRONG"
    print(f"case 2 (short chat not trimmed) -> {result}")

def test_long_history_trimmed() -> None:
    history = [{"role": "system", "content": "be nice"}]
    for number in range(1, 6):
        history.append({"role": "user", "content": str(number) * 400})
    trim_history(history, 250)
    result = "OK"
    if history[0]["role"] != "system":
        result = "WRONG"
    if history[-1]["content"] != "5" * 400:
        result = "WRONG"
    print(f"case 3 (long chat trimmed)     -> {result}")

if __name__ == "__main__":
    test_estimate_tokens()
```

```
    test_short_history_untouched()
    test_long_history_trimmed()
```

```
# tests/unit/test_schemas.py
# Checks the shapes accept good data and reject bad data. Each case prints
# OK or WRONG. No API key needed.
# Run from the repo root: python -m tests.unit.test_schemas
from pydantic import ValidationError

from src.schemas import RepairTicket, RouteDecision

def test_ticket_without_a_name() -> None:
    ticket = RepairTicket(
        customer_name=None, device="laptop", urgency="medium",
        summary="Laptop won't turn on.",
    )
    result = "OK"
    if ticket.customer_name is not None:
        result = "WRONG"
    print(f"case 1 (no name allowed) -> {result}")

def test_ticket_rejects_unknown_device() -> None:
    result = "WRONG"
    try:
        RepairTicket(customer_name="Sara", device="fridge",
                     urgency="low", summary="x")
    except ValidationError:
        result = "OK"
    print(f"case 2 (bad device)      -> {result}")

def test_route_rejects_unknown_agent() -> None:
    result = "WRONG"
    try:
        RouteDecision(agent="billing", reason="looks like billing")
    except ValidationError:
        result = "OK"
    print(f"case 3 (bad agent name)  -> {result}")

if __name__ == "__main__":
    test_ticket_without_a_name()
    test_ticket_rejects_unknown_device()
    test_route_rejects_unknown_agent()
```

```
# tests/integration/test_live_agents.py
# Checks the app against the real OpenAI API. Each case prints OK or WRONG.
```

```
# Needs a valid OPENAI_API_KEY in .env.
# Run from the repo root: python -m tests.integration.test_live_agents
import json

from openai import OpenAI

from src.agents.intake import intake_agent
from src.agents.router import decide
from src.config import load_config
from src.models.llm import (
    check_api_key_at_startup,
    create_client,
    send_message,
)

SAMPLES_PATH = "data/sample/messages.json"

def test_first_reply(client: OpenAI) -> None:
    reply = send_message(client, [], "Hello")
    result = "OK"
    if not reply:
        result = "WRONG"
    print(f"case 1 (first reply) -> {result}")

def test_bad_key() -> None:
    bad_client = OpenAI(api_key="sk-fake-key-123")
    result = "OK"
    if check_api_key_at_startup(bad_client):
        result = "WRONG"
    print(f"case 2 (bad key)     -> {result}")

def load_samples() -> dict:
    with open(SAMPLES_PATH) as samples_file:
        return json.loads(samples_file.read())

def test_every_sample_gets_a_ticket(client: OpenAI, samples: dict) -> None:
    result = "OK"
    for message in samples["intake"]:
        if intake_agent(client, message) is None:
            result = "WRONG"
    print(f"case 3 (tickets)     -> {result}")

def test_no_invented_name(client: OpenAI, samples: dict) -> None:
    ticket = intake_agent(client, samples["intake"][1])
    result = "OK"
    if ticket is None or ticket.customer_name is not None:
        result = "WRONG"
```

```
    print(f"case 4 (no name)     -> {result}")

def test_routing(client: OpenAI, samples: dict) -> None:
    result = "OK"
    for case in samples["routing"]:
        if decide(client, case["message"]).agent != case["expected"]:
            result = "WRONG"
    print(f"case 5 (routing)     -> {result}")

if __name__ == "__main__":
    config = load_config()
    client = create_client(config)
    samples = load_samples()
    test_first_reply(client)
    test_bad_key()
    test_every_sample_gets_a_ticket(client, samples)
    test_no_invented_name(client, samples)
    test_routing(client, samples)
```

Expected unit output (all `OK`):

```
case 1 (missing setting) -> OK
case 2 (empty setting)   -> OK
case 3 (config loads)    -> OK
```

```
case 1 (40 chars = 10 tokens) -> OK
case 2 (short chat not trimmed) -> OK
case 3 (long chat trimmed)     -> OK
```

```
case 1 (no name allowed) -> OK
case 2 (bad device)      -> OK
case 3 (bad agent name)  -> OK
```

That's the whole project. If your version routes a question to the Concierge, turns a problem into a correctly-shaped ticket, leaves an unstated name empty, and never crashes on a bad key or a long message — you've built it.